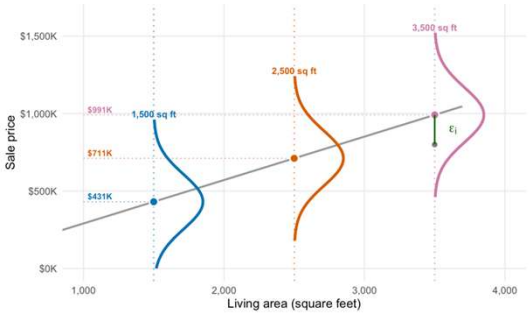
Agenda

- Earlier
  - Simple Line Fitting
  - Linear algebra approach
  - MLE approach
- Today
  - Distribution of the estimators
  - Unknown variance
  - Multiple Linear regression



$$Y_i \sim \mathcal{N}(\alpha + \beta x_i, \sigma^2)$$

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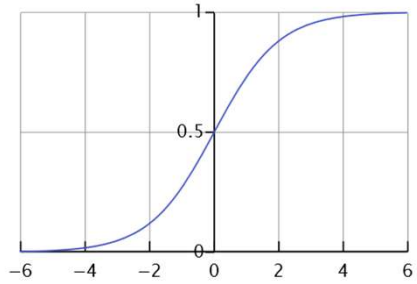
Regression asides

OLS, Logistic, polynomial, multiple

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Logistic Regression

- Consider $\eta = \beta_1 x_1 + \cdots + \beta_k x_k + \alpha$
- Normal linear regression: $\mathbb{E}[Y|x] = \eta$
- Logistic regression: Probability of Y =1 where $\Pr(Y = 1|x) = \sigma(\eta)$
  - $\sigma(\eta) = \frac{1}{1+e^{-\eta}}$
  - Parameter estimates use iterative methods.



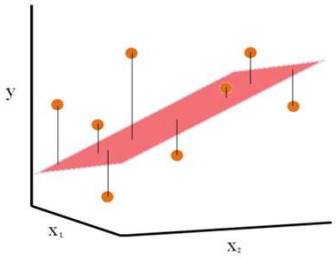
4

Polynomial Regression

- This is still linear in the parameters
- $Y = \beta_1 x + \beta_2 x^2 + \cdots + \beta_r x^r + \alpha + \epsilon$
- OLS treats x, x^2, …, x^r as the predictor columns.

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Multiple Linear Regression



- $Y \mid x_1, \ldots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
- $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
- Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
- $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix}$
- $Y = X\beta + \varepsilon$

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Review: Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,\ldots,X_n$
- **Estimator**: A statistic $T(X_1,\ldots,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter

Oct06

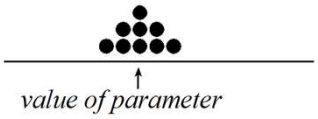
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Understanding the derived parameters

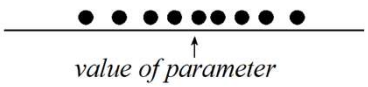
A look at bias, variance and mean-squared error of an estimator

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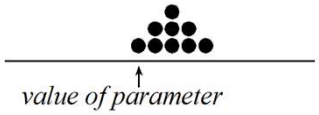
Review: Bias and precision



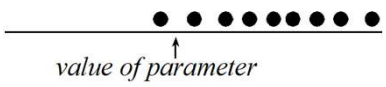
(a) No bias, high precision



(b) No bias, low precision



(c) Biased, high precision



(d) Biased, low precision

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Estimators are random variables

- The sample mean estimator is a random variable
- For independent and identically distributed (iid) random variables with a population mean μ, and finite variance σ^2
  - Distribution only for normal: $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - Sample mean: *unbiased* estimator of population mean because $\mathbb{E}[\bar{X}] = \mu$
  - We also defined the estimator $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$
    - Distribution only for normal: $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$
  - For i.i.d. samples with finite variance: Unbiased estimator of the variance because $\mathbb{E}[S^2] = \sigma^2$

Note the denominator

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ML Estimators

- Maximum likelihood estimation requires a specified statistical model.
  - The model supplies the likelihood used to estimate its parameters
  - If you have no parametric model in mind, then the MLE recipe is probably irrelevant
- The Maximum Likelihood estimator is also a random variable
  - Why? Because its value is a function of the samples from some underlying distribution.
  - The ML estimator has its own sampling distribution, mean and variance.
  - An ML estimate is the observed value of the ML estimator.

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Estimator bias

- The expected value of the estimator, i.e. E($\hat{\theta}$) may not be equal to the true parameter θ, i.e., it may be biased.
  - Define: Bias of the estimator: $E(\hat{\theta}) - \theta$
- Earlier: when given samples are modeled as coming from a normal
  - ML estimator for the mean of a Gaussian is unbiased
  - ML estimator for the variance is biased
  - For large n, the bias is not a major issue

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

Observed μ̂=x̄, with I=n (our usual notation). The random mean estimator is X̄.

Note the denominator

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$

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Estimator variance

- Merely knowing the estimator bias does not give us a complete idea of the quality of the estimator.
  - What if different samples of X gave wildly different estimates of θ for the same estimator?
- The variance of an estimator is (as expected!):

$$\mathrm{Var}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right]$$

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Evaluating a point estimator

- Let X= $(X_1, X_2, \ldots, X_n)$ be a set of n i.i.d. random variables from a given distribution with parameter θ.
- Suppose $\hat{\theta}$ is an estimator of θ
  - $\hat{\theta} = g(X_1, \ldots, X_n)$
  - How does one conclude whether a given estimator is good?
- One metric for this is to evaluate $\left(\hat{\theta} - \theta\right)^2$
  - This is a random variable
  - Mean squared error: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right]$
- We desire estimators with low mean-squared error

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Bias and Variance

- Algebraic identity: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right] + \left(\mathbb{E}[\hat{\theta}] - \theta\right)^2$

variance

Square of bias

- Suppose θ is the population mean we wish to estimate.
  - An estimator that outputs a constant c has a bias $c - \theta$ (zero variance; unbiased only if θ equals c)
  - An estimator that outputs X_i (for some i) will have no bias, but it will inherit the variance of X_i and its variance does not decrease as the sample size increases

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Slope estimator B $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

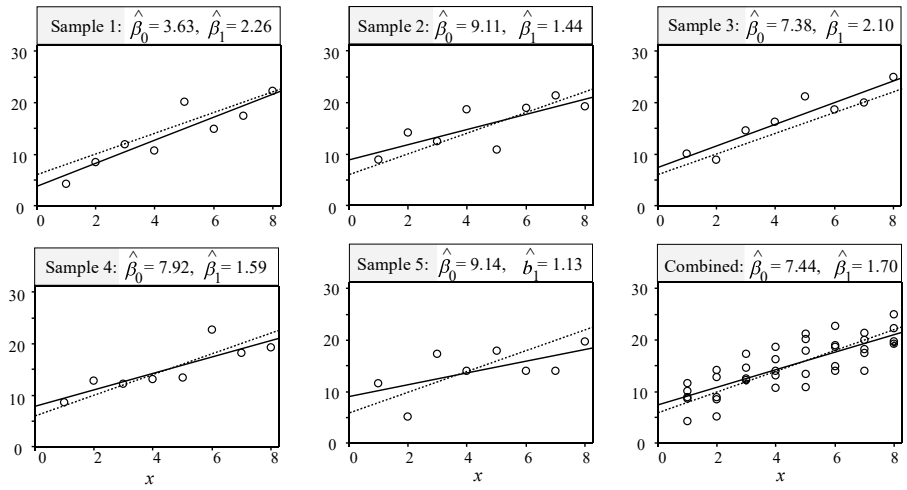
Intercept estimator A $= \bar{Y} - B\bar{x}$

Linear Regression Estimator

Notation from book: The A and B are estimators of parameters α and β in $Y = \alpha + \beta x + \epsilon$. They are random variables

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Example: Sampling variation

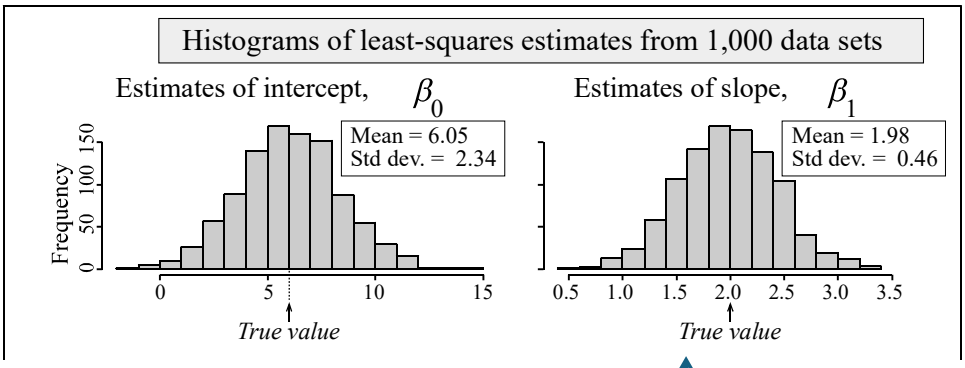


Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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Example: Distribution of estimator



Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

intercept

slope

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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Linear regression Parameters

- estimate $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}, Y_i = \alpha + \beta x_i + \varepsilon_i, \varepsilon_i \sim N(0,\sigma^2)$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \Longrightarrow B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{j=1}^{n}(x_j-\bar{x})^2}$ ← Note the denominator
  - But $\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y}) = \sum_{i=1}^{n}(x_i-\bar{x})Y_i - \bar{Y}\sum_{i=1}^{n}(x_i-\bar{x})$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})Y_i}{\sum_{j=1}^{n}(x_j-\bar{x})^2} = \sum_{i=1}^{n}\frac{(x_i-\bar{x})}{\sum_{j=1}^{n}x_j-\bar{x})^2}Y_i$ (a linear combination of Y_i)
- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$

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Linear regression Parameters

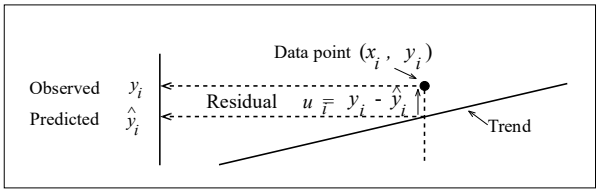
- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$
  - Unbiased estimator: $\mathbb{E}[B] = \beta$
- $A = \bar{Y} - B\bar{x}, A \sim \mathcal{N}(\alpha, \mathrm{Var}(A))$
  - Unbiased estimator: $\mathbb{E}[A] = \alpha$ and $\mathrm{Var}(A) = \frac{\sigma^2 \sum_i x_i^2}{n(\sum_i x_i^2 - n\bar{x}^2)}$

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Linear regression Parameters

- Define $\mathrm{SSR} = \sum_{i=1}^{n}(Y_i - A - B\mathrm{x}_i)^2$
  - Then $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}, \Longrightarrow \mathbb{E}[\frac{SS_R}{\sigma^2}] = n - 2$
  - Unbiased: $\mathbb{E}\left[\frac{\mathrm{SSR}}{n-2}\right] = \sigma^2$

Note the denominator



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Least-squares fitting as MLE

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
  - Minimizing the negative log likelihood results in
    - Same expression for A and B as in the ordinary least square (linear algebra)
    - Different expression for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$

Note the denominator

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Review

- OLS and Gaussian MLE give the same coefficient estimators A and B.
- σ^2 is estimated from the residuals in both cases. They differ only in the divisor: n (MLE) or n−2 (unbiased).
  - The n−2 is the degrees of freedom left after fitting 2 parameters, the direct generalization of n−1.
- Normality buys exact sampling distributions and inference, not the point estimates.
- For large n the divisor difference is small. Use S^2=SSR/(n−2); software reports its observed value s^2.

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